First incompleteness theorem for omega-consistent theories

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Editorial note. Both cited lemmas are corpus cards, so the composition is corpus-internal end to end.

Editorial scope. One original theorem of the unit, counted once: the omega-consistent form of the first incompleteness theorem. The counted theorem and its author proof are retained; both cited lemmas are corpus cards.

Allowed standard prerequisites (editorial).

- The Godel sentence G_T for an axiomatizable theory T extending Q, as constructed in the unit's incompleteness section.
- Omega-consistency of T: T does not prove both an existential statement and the negation of each numeral instance of its matrix.
- The two sibling results proved in the same unit: a consistent T does not derive G_T, and an omega-consistent T does not derive the negation of G_T.
- The one-line observation that omega-consistency implies consistency: an inconsistent theory derives every numeral instance of any formula together with the negated existential, so it is omega-inconsistent.

Source reference. Original lemma cons-G-unprov: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 455–455. The proof cites the unit's lemma that a consistent T does not derive its Godel sentence. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. Published as corpus card X2-incompleteness-incompleteness-provability-cons-G-unprov.

Source reference. Original lemma omega-cons-G-unref: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 488–488. The proof cites the unit's lemma that an omega-consistent T does not derive the negation of its Godel sentence. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. Published as corpus card X2-incompleteness-incompleteness-provability-omega-cons-G-unref.

Theorem 0.1. *Let* ThT *be any* ω*-consistent, axiomatizable theory extending* ThQ*. Then* ThT *is not complete.*

Proof. If ThT is ω-consistent, it is consistent, so Th$T \vdash /G_{\mathrm{Th}}T$ by cons-G-unprov . By omega-cons-G-unref , Th$T \vdash /\neg G_{\mathrm{Th}}T$. This means that Th$T$ is incomplete, since it derives neither $G_{\mathrm{Th}}T$ nor $\neg G_{\mathrm{Th}}T$. □